

Notification System Refactor Guide

Purpose

This document describes the refactoring applied to the backend notification system to improve control, reduce unnecessary notifications, and introduce role-based filtering.

Overview

- Prevent unnecessary notifications (e.g., login spam)
- Enforce role-based notification delivery
- Improve overall flow and maintainability

Role-Based Notification Filtering

A filtering function was introduced: `is_notification_allowed(user, category)`

Behavior

- Students only receive: `event_reminder`, `attendance`, `missed_events`, `low_attendance`
- Admins are allowed to receive all notification categories
- Other roles are skipped by default

Notification Flow Update

The main notification function `send_notification_to_user(...)` now includes an early exit condition to skip disallowed notifications.

Effect

- Prevents unnecessary processing
- Avoids sending irrelevant notifications
- Improves system efficiency

Account Security Notification Control

The function `send_account_security_notification(...)` now sends notifications only when user preferences allow it and the event is marked as suspicious.

Notification Channels

- In-app notifications
- Email notifications
- SMS notifications (placeholder)

Each channel logs its result using `create_notification_log(...)`.

Logging Behavior

- Status: sent, failed, or skipped
- Channel: email, sms, in_app, or none

This ensures traceability and debugging support.

No Changes Applied To

- email_service/ (email delivery system)
- alembic/ and migrations/ (database history)
- config.py (core configuration)

Summary

- More controlled
- Role-aware
- Less prone to spam
- Easier to maintain

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